AI in Human Resources & Talent Management

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Introduction and Case Context:

An organization's most valuable resource is not always the technology, products, or customers, but the people who implement the strategy and make it work. Managing human resources is becoming increasingly challenging, as organizations operate in an environment of a high volume of applications, competition for talent, evolving employee expectations, and the need to address concerns while maintaining productivity. In particular, it is critical to attract and motivate professionals in the IT industry, as most companies' performance depends on the employees' expertise.

The hypothetical company mentioned in this report operates in India and is a large-scale IT organization. It experiences three main human resource issues, including high turnover, difficulties in the hiring process, and varying levels of employee engagement and development. These problems impact the organization's performance, as well as the employees' experience and productivity.

Human resource management practices are essential in any organization, as they directly affect the employees' motivation and productivity. However, the traditional methods of recruitment and engagement are no longer sufficient due to the complexity of modern problems. Artificial intelligence can help resolve some of the issues by analyzing the data, engaging employees in a meaningful dialogue, and automating the processes. However, the technology can produce biased results when given faulty instructions, making it critical to acknowledge employees' needs.

This report proposes an AI-enabled Human Resource Management (HRM) system to resolve the issues faced by the company. The system will address the concerns of recruitment, attrition, learning, and engagement, while also acknowledging the importance of human input in the decision-making process.

AI-Supported Recruitment and Fair Hiring:

Recruitment is one of the first processes where AI can be implemented to benefit the organization. It is time-consuming and tedious for the recruiter to scan through thousands of resumes for a single vacancy, particularly when technical expertise, soft skills, and work ethic are required. The hiring professional will also face challenges when explaining their rationale to the candidates or the employees, as some of the decisions are subjective by nature.

The AI-enabled recruitment system, as proposed in this report, will be based on objective and just criteria. Before implementing the technology, the recruiters and managers should discuss and define the most desirable qualifications, experience, skills, and competencies. They should also consider the responsibilities and requirements of the vacant position and how they align with the applicant's working conditions and opportunities for development. For instance, the hiring managers should define the criteria for a data analyst vacancy, such as the required SQL and Excel skills, analytical abilities, and communication competencies. It will enable the AI system to organize the applications according to the candidates' qualifications and competencies instead of relying on assumptions, keywords, and subjective evaluations.

It is also vital to understand that AI algorithms cannot resolve disagreements or biases unless they recognize them. Thus, the system will be utilized to arrange the applications and highlight the candidates' qualifications, experiences, and skills relevant to the vacant position. For instance, if a hiring manager receives a resume from an applicant who identifies as a 'Python programmer,' the application will be sorted under the 'data analyst' category since programming languages such as Python are often used in data processing and analysis. The technology will also assist in the initial screening of the applicants, acknowledging the differences in their profiles. As a result, the system could be integrated with other aspects of the recruitment process, such as interview scheduling and candidate engagement and support automated confirmations and information retrieval functions.

However, it is inadvisable to rely solely on technology to make recruitment decisions, as the process involves several subjective and sensitive factors. If the company utilizes an AI system to make recruitment decisions based on the data of the past hires, the outcomes might be unfair or even illegal. The technology might perpetuate past issues if the recruiters and managers have historically given priority to candidates from a specific university or those with certain extracurricular activities or hobbies. In contrast, the candidates without such recommendations from the AI system might be disadvantaged despite their qualifications. It is

essential to use the system to organize the applicants according to their merits and eliminate the factors that do not align with the vacancy requirements.

The system can be utilized to sort, arrange, and support the decision-making process, but it should not be the sole basis of the final judgment. For instance, if a hiring manager uses an AI system to determine whether a candidate is suitable for the technical writer position, they should be able to override the recommendation if they believe the application does not align with the company's requirements. The managers should also ensure that the system does not use irrelevant information, such as the applicants' hobbies, to screen out unqualified candidates or Favor one group over the other. The AI system can be utilized to sort the applications, shortlist the candidates, and help the hirers identify the best choice based on the evidence. The final hiring decision should be made by the recruiters or the hiring managers, who can review the evidence provided by the system and consider other career-related information not accounted for by the AI model.

The managers and HR specialists should also ensure that the candidates have recourse if they are not satisfied with the outcome of the recruitment process. They should be able to review their application and the evidence used by the system to make decisions. If the model made a recommendation based on irrelevant or unlawful criteria, the hirers should be prepared to revise their decision. The recruitment system should also be tested to ensure that it does not sort the candidates differently based on protected categories such as race, religion, gender, or ethnicity. If such data is available, the managers should be able to delete it before the final decision is made. The hiring managers should ensure that the model does not overlook essential information, such as the candidate's relevant work experience, or include unnecessary data, such as the applicant's age, in the final recommendation.

Thus, an AI recruitment system can significantly enhance the hiring process by short-listing the candidates and providing the hirers with objective evidence to make the final decision. It is essential to ensure that the system is used ethically and does not replace the human input in the recruitment process.

The model should be applied to the most time-consuming aspects of the recruitment process. As a result, the system is likely to improve the quality of hiring decisions while also eliminating biases and discrimination.

Predictive Attrition Analytics and Employee Retention:

High turnover is another significant issue facing the organization. It impacts the productivity of the current employees while also increasing the financial strain on the company as it searches for suitable replacements. Losing an employee also means that the organization might have to train a new worker and might lose the productivity and knowledge of the departing worker, particularly when the vacant position is not easily filled. This issue is exacerbated when the employees who have worked together leave at the same time, impacting the productivity of the entire team. While some employees will always leave to seek better opportunities, such losses can be avoided when the hirers and managers utilize technology to predict and mitigate attrition.

The attrition system will focus on the candidates' most relevant information, including their length of service, job rotation and transfers, training and development, workload, and promotions. The information is essential because it can impact the employee's productivity and willingness to continue working with the organization. Other relevant evidence includes voluntary separation surveys, workload data, and the employees' suggestions and concerns. The data should be analyzed by AI to identify the patterns that contribute to an employee's decision to leave the organization. The information should not include irrelevant or protected categories such as the employee's ethnicity, age, gender, or hobbies.

The attrition analytics tool will focus on the employees' most relevant data to identify the likelihood that they will leave within the next year. It will make recommendations based on the evidence. However, such information should be used as a reference rather than a definitive statement that an employee will leave. There are too many subjective and objective factors that contribute to an employee's decision to quit, and the AI model will only be able to highlight the objective reasons, such as a lack of training, mentorship, or compensation, which the organization can address. As mentioned above, leaving is a subjective decision that depends on an employee's circumstances, motivations, and personal life.

Thus, the system should not be utilized to label employees as a 'flight risk.' It is essential to use the attrition analytics tool to make evidence-based recommendations to improve the retention rate. If the model indicates that employees in a specific department are more likely to leave, the human resource managers can review the

evidence and make necessary adjustments to improve the retention rate in that department. For instance, if the attrition analytics tool indicates that the employees in a specific team are likely to quit, the HR managers can work with the managers in that department to ensure that the employees have adequate support, opportunities for advancement, and training.

The attrition analytics tool should also be used to hold discussions with the employees about their concerns and future in the organization. It is essential for the managers and HR specialists to hold onto their power and not rely solely on data when making decisions about employees' futures. Once the AI model has made recommendations based on the evidence, the managers should hold a meeting with the employees to discuss the issue. It is also vital to ensure that the employees are aware of the system. They should know that the evidence is based on facts and not assumptions to build trust in the process. The system can be utilized to keep track of the employees' concerns and address them accordingly. For instance, if the attrition analytics tool indicates that an employee is likely to quit, the managers can use the information to hold a motivational meeting with the worker.

An attrition analytics tool should be utilized to generate evidence for human resource managers to make decisions. It is critical to ensure that the tool is not used to make unilateral decisions about the employees' futures. It is also essential for the system to highlight that the employees' subjective motivations are difficult to interpret, and the most effective way to address concerns is through direct communication with each employee.

Personalized Employee Development and Career Progression:

An organization's employees need to acquire the relevant skills to perform their jobs efficiently. This is particularly true for the IT industry, where the needs of the clients and the operations of the business change at a rapid pace. However, not all employees will have the same proficiencies, and some will require more training and mentorship than others. Furthermore, providing training and development programs tailored to each employee's needs is a waste of the company's resources if they do not intend to utilize those employees in the future. Thus, it is essential to ensure that the employees have the relevant skills for the current vacancies while also preparing them for future positions.

AI can be utilized to ensure that the employees receive the necessary training and mentorship while also supporting the managers in making decisions about the employees' future with the organization. The AI model will have access to the employees' skills, training, and work ethic, as well as the vacant positions and the required proficiencies to perform those jobs successfully. It will make recommendations about the training and development programs based on the employees' interests and the company's needs. For instance, the model can suggest that an employee working as an application specialist acquire the skills of a data analyst to fill a vacant position in that department. The AI model will make recommendations based on the evidence, and the employees will decide whether they want to pursue the suggested training programs and subsequent positions.

Thus, AI can transform the employees' learning and development while also supporting the human resource managers in training the workers and preparing them for future roles in the organization. It is vital to ensure that the recommendations made by the AI model are relevant and based on objective evidence. The information should also be accessible to the employees so that they feel involved in the process and are aware of the evidence that the managers will use to make decisions about their future. Individuals who complete the training and development programs should be able to apply the newly acquired skills in the workplace, and the managers should ensure that the employees receive adequate mentorship and encouragement to use those proficiencies.

AI-Driven Employee Engagement:

There are multiple factors that influence employee engagement, including leadership, relationships, recognition, workload, and opportunities for advancement. An employee's engagement also depends on the job role, years of service, and the employee's ability to influence processes and decisions within the organization. Employees are also likely to be engaged when they receive support from their colleagues and superiors, have opportunities to develop their skills, and are recognized and rewarded for their hard work.

However, it is difficult to measure employee engagement, as it is subjective and influenced by several variables. An employee might be highly engaged at the beginning of their career but become demotivated as they advance to a more senior position. This lack of engagement is difficult to quantify as most traditional methods of measuring employee engagement are numerical scores based on the results of a single survey. Thus, the employee's engagement level cannot be determined by a single number or a set of questions designed to measure their current attitude toward the job, regardless of the role and the years of service.

The AI model can be utilized to make recommendations based on the evidence provided by the employees. Most businesses ask for employee feedback through exit interviews, suggestion boxes, and engagement surveys. The AI model can be used to organize this information and identify the most pressing concerns of the employees. It should be able to separate the issues that the employees have raised and determine whether they require the immediate attention of the human resource managers. It will also highlight the topics that the majority of employees are concerned about, such as career progression, workload, and recognition. For instance, if the majority of the employees who have left the organization cited the lack of clarity about the promotion criteria as the primary reason for their decision, the human resource managers should review the promotion policies and ensure that they are transparent and accessible to all employees.

Thus, an AI-driven employee engagement system can transform the relationship between the employees and the organization by giving the workers a voice and making their concerns a priority. It is essential to ensure that the recommendations made by the AI model are relevant and based on objective evidence. The employees should also be able to provide feedback to the system so that it can continue to learn and make better recommendations. Furthermore, the employees should be aware that their concerns will be addressed by the human resource managers rather than the technology. The process will allow the workers to remain engaged and motivated, knowing that their input has been considered and that the company is willing to make relevant adjustments rather than ignoring their concerns.

The technology can be utilized to make recommendations based on the evidence provided by the employees regarding their concerns and opinions about the workplace environment. This information should be analyzed by the AI system and transformed into digestible insights for the human resource managers. The managers should then make the necessary adjustments to improve the employees' experience and encourage them to remain in the workplace. For instance, if the majority of the employees have indicated that they are dissatisfied with their current workload and it impacts their work-life balance, the managers should adjust the project deadlines to accommodate the employees' needs. A similar action should be taken if the employees indicate that they do not have access to the necessary training and development programs to improve their performance or if they feel that their contributions are not recognized and rewarded sufficiently.

The AI-driven employee engagement system can also be utilized to identify the employees' needs and make recommendations regarding the most relevant recognition programs. The employees' feedback and concerns should be considered, and the most pressing issues should be addressed first. However, the technology should only be used as a tool to guide the human resource managers and not to make unilateral decisions about the employees' future or the organization's needs.

AI-assisted engagement analysis should focus primarily on group-level themes and organizational conditions. It should not attempt to read an employee's emotions, mental health, loyalty or attitude from their facial expressions, voice or private communication. This can be challenging to accurately interpret, intrusive to an employee's personal life and difficult to contest. Where individual support is required, it should generally start from regular, respectful discussions and established employee support channels rather than an automated emotional analysis.

Most importantly, the feedback needs to lead to action. If employees regularly voice concerns which are not visible in the results, their confidence and trust in the survey process may be impacted. Where applicable, HR should communicate the themes identified and share which actions the company intends to take, as well as the progress. Not all suggestions can necessarily be incorporated, but employees must feel that their feedback has been thoroughly reviewed and considered in a process they can understand. AI can assist the identification of concerns, but trust is established through how the organization reacts.

The company can measure the area through the participation rates, employees' perception of communication and fairness, resolution of commonly-voiced concerns, qualitative analysis of the actions taken based on the

feedback. It should interpret the findings together, rather than attempting to summarize the engagement into a single metric. The focus should be on improving the conditions and addressing the concerns impacting the employee experience, not merely measuring it more often.

Proposed Integrated AI Solution:

The four proposed applications (recruitment, attrition analytics, employee development and engagement) should be designed as connected elements of an integrated AI-supported solution for Human Resource Management (HRM). It would enable reduced data entry and consistent reporting across use cases, as well as broader analysis and management insights by linking relevant processes. At the same time, integration should not necessarily mean that any employee record is accessible to any system and user. Access to employee information should be limited to the processes actually required for a particular analyst function. Figure A1 in Appendix A provides an overview of an integrated AI-supported HRM framework.

The HR data layer will aggregate relevant employee information from recruitment systems, the HR information system, learning databases, internal mobility records and voluntary employee feedback. The information should be validated for accuracy, relevance and completeness prior to being used for analysis and reporting. The company should document the sources of each data element, its purpose, access and retention requirements. Where irrelevant or sensitive, employee information should not be gathered for mere possibility of analysis.

The AI and analytics layer will host relevant models which can analyze the data to address specific HR questions. Recruitment analytics can match the required skills with the applicant pool, while attrition analytics can identify patterns associated with each departure. Learning analytics can recommend individual development paths based on employees' skill requirements and career interests, while text analysis can organize voluntary employee feedback into common themes. Each model should be developed with a specific purpose in mind and carefully reviewed in terms of consequences of its application. A recruitment matching tool, for example, would be significantly impacted by different types of errors than an attrition model and require different considerations.

The HR and manager decision-support layer would facilitate employee analysis and decision-making by providing a relevant, accessible view on the information and models available. An overview dashboard, for example, can provide a summary of the recruitment pipeline, skills gaps, mobility trends or attrition patterns. The same dashboard can give a manager or HR business partner an insight into their team's requirements or concerns. Individual-level employee analysis should be limited to relevant decisions and supported by appropriate safeguards and controls. The dashboards and reports should explain the indicators and their implications, as well as the actions, if any, that the audience can take.

The employee access and support layer will ensure that employees have the opportunity to access relevant learning recommendations, view internal mobility opportunities, provide voluntary feedback and understand how their information is used in the system. They should be able to correct factual inaccuracies where possible, as well as request a human review of an individual consideration or action. The system should offer a way for employees to raise concerns and questions about the application of AI in their specific case.

The company should not rely on a unified employee score which would attempt to encapsulate an individual's value, loyalty, performance, engagement and potential. It would rely on different dimensions with potentially varying levels of importance and risk, and should therefore be separated into distinct tools. An employee may have significant shortcomings in one area while excelling in others, but a single metric would represent these differences as comparable values. Separation is also necessary to ensure that employees are not penalized for factors beyond their control or unfairly advantaged over competitors.

Integration should enable a holistic view on the workforce without losing the context around individual facts and figures. A group trend should not automatically be interpreted as a fact, but should be used to explore options for improvement in collaboration with employees. If an attrition analytics tool, for example, identifies a pattern in one team which correlates with a lack of skills, internal mobility opportunities and employee learning, it can inform the decision to investigate these factors. But it should not automatically assume that the lack of training causes the turnover and disregard the employees' perspectives on the matter.

Implementation Strategy:

Define the problem and establish governance:

The initial phase should focus on understanding the company's requirements and determining the problems which require an AI-supported solution. HR leaders, business managers, data specialists, legal or compliance representatives and employee representatives (where appropriate) should jointly define the objectives. The company should document the intended application of each tool, the decisions it will support, the information it will use and the aspects which will continue to rely on human judgment. At the same time, the company should assess the suitability and quality of the information which will be used for analysis and modeling. The historical records may contain inconsistencies, irrelevant data points, incorrect or inconsistent labeling and other issues which should be addressed prior to preparation and use.

Run a limited pilot:

The organization should identify a single use case for a pilot, such as AI-assisted skill matching for a limited vacancy set, or a learning recommendation pilot for one job family. This will reduce the risks and resource commitments while still providing valuable insights into the practical application of the tool. It will also enable the comparison of results and user experience with the existing process. The pilot should include clearly-defined success and risk assessment metrics. For recruitment, they can include the time spent on initial screening, the relevance of shortlisted candidates, candidate feedback and comparative results across similar groups. For learning, they can focus on employee participation, completion of relevant learning units, practical application of new skills and employee feedback. The company should also track the emergence of concerns, complaints or instances where the system did not provide a useful recommendation.

Evaluate and improve:

The company should analyze the results of the pilot in terms of effectiveness and risk. A tool which reduces the time spent on a task may still introduce new concerns if it provides unactionable recommendations or eliminates relevant candidates with no possibility of review. Similarly, a learning recommendation system may generate numerous suggestions, but these can offer limited value if the employees do not have the capacity to engage with them. The evaluation should therefore consider the impacts on the business, employees and fairness of outcomes, as well as the quality of human oversight. Where the results are unsatisfactory, the company should reconsider the model, process, data or even abandon the initiative if no sufficient improvements can be identified. It should not continue to invest resources in a solution which has demonstrated unacceptable results.

Expand with appropriate controls:

The company should only look to expand the application or audience of a tool after a successful pilot and risk assessment. The expansion should be accompanied by appropriate training for the new users. They should understand the tool's purpose, data sources, analytical outputs and limitations. The users should also be aware that an AI-generated recommendation is not a fact, but a proposal which requires individual consideration. The tool should not replace an employee's or manager's judgment, but support it based on the available information.

Monitor continuously:

AI tools and models can become less accurate as the business environment changes. The company should therefore continuously review the performance of the models, employee concerns, data quality and fairness impacts at regular intervals. Significant changes to the data, implementation instructions or intended use should go through a separate evaluation before being implemented. The employees should always have an option to raise a concern about an individual case.

Evaluation and continuous monitoring are essential parts of responsible AI implementation, which should not be viewed as a one-time project.

Ethical Considerations, Risks and Limitations:

The application of AI in HR presents unique risks, as the potential outcomes can impact employees' careers and access to opportunities. The company should therefore consider ethical governance not as a final checkpoint, but an integral part of design and implementation.

Fairness and avoidance of bias are of particular importance, as the use of biased variables can undermine the trust of employees and lead to poor business outcomes. While the use of a sensitive variable (such as race) for decision-making is clearly unacceptable, other variables can indirectly impact the results by reflecting existing biases. Fairness should therefore be achieved by comprehensive review of all variables used for analysis, outcome validation across relevant groups (where lawful and possible) and ongoing monitoring of changes.

Similarly, employees' privacy and data protection should be considered at all stages. An employee record can contain numerous sensitive elements which should only be viewed and processed by those who are entitled to make a decision. The company should limit the collection of employee data to the information which is relevant to the purpose and authorized by the employees. It should also ensure the confidentiality of this information, as well as the ability to delete or restrict it where appropriate. The employees should be informed of the data which is used for processing, as well as its impact on their experience with the system. Data collected for one purpose should not be used for unrelated investigations without the separate consent of the employee.

Transparency and explainability are essential for trust, accountability and effective employee experience. Where a candidate is not shortlisted, or an employee receives a specific recommendation, the organization should be able to explain the rationale behind the outcome in terms of relevant factors. A black box approach will limit the ability of HR professionals and employees to contest the result and investigate concerns. The company should provide the documentation and analysis which can support the recommendation at an appropriate level (for example, a manager does not need to know the detailed mechanics of a model, but should understand the factors which impact the result). Transparency should not necessarily mean full disclosure of all details to all users, but should focus on the information which impacts their decision-making.

Human oversight should be integral to the process and not merely exist for compliance. A manager who approves an AI recommendation without additional screening will generally not have performed their duty adequately. HR professionals should have the authority and knowledge to review the recommendation, request additional information, and reject it if it is not suitable for the case. Decisions with significant impact on employees' careers should rely on careful human consideration.

Employee trust and acceptance can be essential for the practical success of the initiative. A system which employees believe will monitor and control them can lead to undesirable behaviors and outcomes. The company should therefore seek to build employee understanding and engagement, rather than focusing solely on the benefits for the business. It should provide the context for the initiative, allow employees to raise concerns and involve them in the process where possible. This will also help identify areas of concern and impact which the implementing team may not have considered.

The limitations of models and reduction of human judgment should be acknowledged. An AI system will generally identify patterns, but will not necessarily understand the context and nuance of individual cases. A manager should therefore have the ability to review and override an algorithm's suggestion, particularly where there is insufficient data or a known flaw in the system. For example, an attrition model may suggest that employees are likely to leave due to a specific factor, but it does not account for the employees' intent to retire. Predictive models should not solely rely on historical data if the business circumstances or employee preferences are known to change.

Finally, the financial and practical costs should be reviewed. AI technology, data analysis and governance require significant resources. If the company's current practices are lacking in relevant areas (for example, employee feedback management), adding an additional layer of complexity may fail to address the key concerns. The company should consider its readiness and determine whether the expected benefits outweigh the costs. If the organization's data and employee processes are currently inadequate, it should dedicate resources to improving its fundamentals rather than immediate AI adoption.

Recommendations:

Start with clearly-defined problems: The company should identify the specific issue it wishes to address (recruitment, retention, development, engagement) and determine the requirements for an AI-supported solution. The business need should shape the choice of the technology, rather than the opposite.

Keep recruitment criteria job-related: AI can support the assessment of the required skills, but structured evaluation and appropriate human judgment should remain an integral part of the recruitment process.

Use attrition analytics to make changes: Predictive attrition modeling should inform the improvement of the employee experience, rather than simply identifying "at-risk" employees and taking disciplinary or financial action.

Use employee feedback responsibly: The company should analyze the voluntarily-provided feedback to identify common themes, but ensure employee privacy and avoid intrusive monitoring.

Establish human accountability: The company should ensure that the limitations of AI are understood and that human judgment remains an integral, accountable part of the process.

Evaluate results: The company should measure efficiency gains and the quality of results, employee experience and long-term business impacts, rather than focusing solely on cost reductions.

Establish employee trust: The company should inform the employees of the purpose of the system, let them know of their options for feedback, and ensure that their concerns can be heard and addressed.

Conclusion:

The true value of AI in Human Resource Management lies not in automating every aspect of the employee lifecycle, but in enabling the organization to make superior judgments. It has the potential to support the company in discovering important patterns, engaging employees and taking timely action to improve the experience and performance of the workforce. For the hypothetical IT company described in this report, AI-assisted recruitment could make the recruiters more efficient and consistent in finding suitable candidates. Attrition analytics could help the company proactively identify and address the root causes of employee departure. Personalized development recommendations could introduce employees to new opportunities and support them in advancing their career. Finally, engagement feedback analysis could help the company understand the concerns which prevent the employees from enjoying their work with the business.

But none of these benefits would be available in isolation, and the tools should be selected and implemented based on the specific needs of the business. A recruitment model is of limited use if the selection criteria are not clearly defined. An attrition model will generally reflect the choices and behaviours of the employees, but it will not understand their motivations and provide actionable guidance. Learning recommendations will only be useful if the business has the capacity to support the employees in furthering their development. An engagement dashboard will motivate and reassure employees only if the company acts on their feedback. These limitations are why the organization should not rely on AI to be an end in itself, but rather use it as a means of supporting its human capabilities. For the company in our scenario, the responsible implementation of AI requires a human-centric approach rooted in understanding the business requirements and addressing the most pressing problems. It should make decisions based on the evidence and ensure that the data underpinning its conclusions is relevant, reliable and valid. The company should apply its initiatives on a limited scale and validate their effectiveness and risks prior to broader adoption. It should involve HR professionals and employees in the process and ensure that critical decisions retain human judgment. These considerations will support the organization in achieving the benefits of AI while minimizing its risks.

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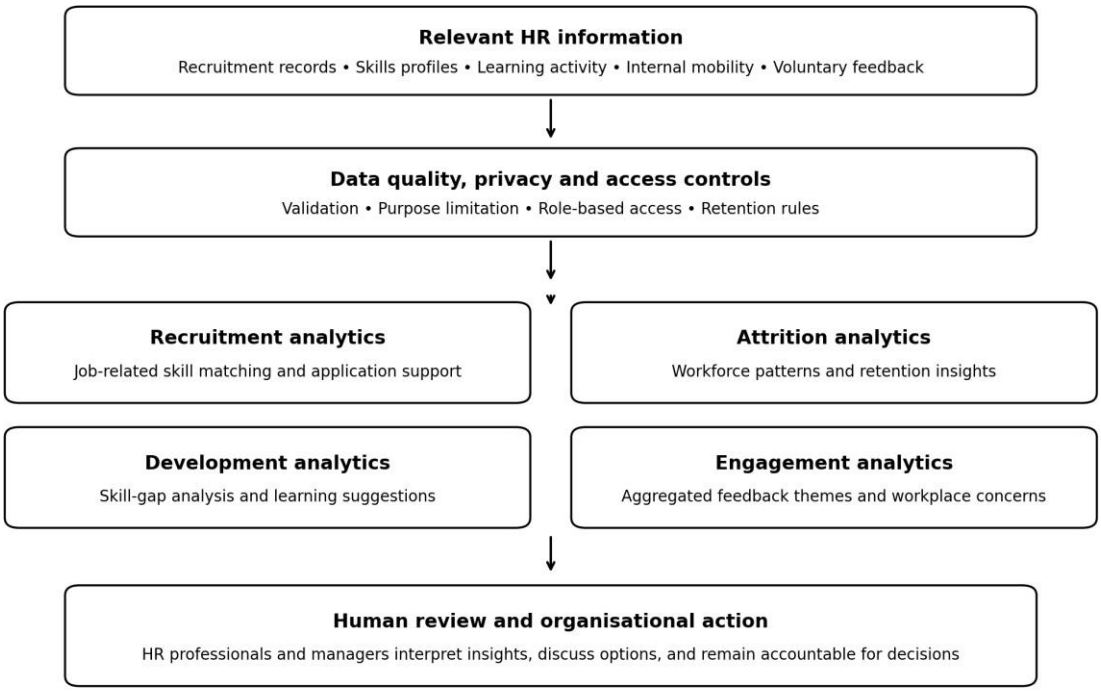
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Appendix A: Proposed Integrated HR System Flow



AI supports HR judgement; it does not replace accountable human decision-making.

Appendix B: Proposed AI-supported HR system architecture

HR area	Purpose	Illustrative inputs	Key safeguard
Recruitment	Support job-related candidate matching and organize applications	Job criteria, skills and relevant experience	Structured criteria, fairness checks and recruiter review
Attrition	Identify workforce patterns that may need retention action	Tenure, role changes, internal mobility and appropriate feedback	Treat outputs as signals; avoid punitive individual labels
Development	Suggest relevant learning and internal career options	Skills profiles, learning history and employee career interests	Keep recommendations optional and connect them to real opportunities
Engagement	Identify recurring themes in employee feedback	Voluntary surveys, exit feedback and suggestion channels	Prefer aggregated themes; avoid intrusive employee surveillance

Appendix C: Integrated HR Decision Framework

This framework would unify the four human resources applications by a common governance process but maintain a distinct decision-making and data access scope for each of them, which would not provide an aggregated employee score.

HR application	AI-supported insight	Human-led response
Recruitment	Job-related skill matches and supporting evidence	Recruiter checks suitability, structured assessments and fairness before a hiring decision
Attrition	Aggregated turnover patterns by role, team or tenure	HR investigates workload, mobility and management conditions before designing retention actions
Development	Skill gaps, employee interests and potential learning pathways	Employee and manager select realistic learning, projects and mobility opportunities
Engagement	Recurring themes in voluntary feedback	HR validates themes, agrees corrective actions and communicates progress

Decision sequence: define the HR problem, validate relevant data, generate a purpose-limited insight, review context and fairness, take an accountable human decision, evaluate outcomes and employee feedback.

Appendix D: Practical Performance Measurement Plan

The following measures are suggested for a pilot and later review. These include baselines, targets and results that must be determined from company data; no improvement is assumed.

Area / indicator	How to measure	How to evaluate
Hiring quality	New-hire performance against defined role criteria, plus retention after joining	Compare cohorts and candidate experience; review differences across groups
Voluntary turnover	Voluntary exits ÷ average headcount × 100 for a defined period	Compare similar roles and periods; consider labour-market and business changes
Internal mobility	Employees moving to another internal role ÷ eligible workforce × 100	Assess whether moves reflect genuine opportunity, not forced transfers
Learning application	Share of trained employees demonstrating relevant skills through work samples or projects	Use manager and employee evidence, not course completion alone
Employee trust	Anonymous survey items on fairness, privacy, clarity and confidence in HR processes	Track themes and response rates; report actions taken without identifying individuals

Evaluation approach: The initial step is determining the definitions and baseline before the pilot, comparing it to the existing process (where possible), and assessing fairness and qualitative feedback alongside efficiency, and investigating external factors preceding any changes to AI.